motionref1 奖励与观测公式

commit `11fe46d` $\quad \Delta t = 0.02\,\mathrm{s} \quad v_z^{\mathrm{cmd}} = 0.01395\,\mathrm{m/s} \quad$ 总奖励 $R_t = \sum_i w_i\, r_i(s_t)\, \Delta t$

1 DeepMimic 模仿项（exp 核，参考量 $\hat{\cdot}$ 取自动作参考 clip 相位帧 k）

$$r_{\mathrm{jpos}} = \exp\Big(-\tfrac{1}{\sigma^2}\, \mathrm{mean}_j \big(\hat{q}_j - q_j\big)^2 \Big), \qquad \sigma = 0.2 \text{ rad} \;\; (w = 2.0)$$

$$r_{\mathrm{jvel}} = \exp\Big(-\tfrac{1}{\sigma^2}\, \mathrm{mean}_j \big(\hat{\dot{q}}_j - \dot{q}_j\big)^2 \Big), \qquad \sigma = 1.5 \text{ rad/s} \;\; (w = 0.5)$$

$$r_{\mathrm{ee}} = \exp\Big(-\tfrac{1}{\sigma^2}\, \mathrm{mean}_h \big\|\hat{\boldsymbol{p}}_h^{\,r} - \boldsymbol{p}_h^{\,r}\big\|^2 \Big), \qquad \sigma = 0.1 \text{ m} \;\; (w = 1.5)$$

$$r_{\mathrm{ori}} = \exp\Big(-\tfrac{1}{\sigma^2}\, \theta_{\mathrm{geo}}\big(\hat{q}^{\mathrm{root}}, q^{\mathrm{root}}\big)^2 \Big), \qquad \sigma = 0.3 \text{ rad} \;\; (w = 0.5)$$

$\boldsymbol{p}_h^{\,r}$：EE 在**根系**的位置（双侧根相对 $\Rightarrow$ 只塑形肢体几何，不锚定全局轨迹）；θ_{geo}：四元数测地角误差。

2　相位啮合调度与爬升收入

$$r_{\text{mismatch}} = \sum_{h=1}^{4} \mathbf{1}[\hat{e}_h(k) \neq e_h] \quad (w = -0.25)$$

$$r_{\text{height}} = \text{clip}\left(\frac{\left(\max(z^{\text{best}},\, z_t) - z^{\text{best}}\right)}{v_z^{\text{cmd}}\, \Delta t},\ 0,\ 12 \right), \qquad z^{\text{best}} \leftarrow \max(z^{\text{best}}, z_t) \quad (w = 2.0)$$

$$r_{\text{signed}} = \text{clip}\left(\frac{z_t - z_{t-1}}{v_z^{\text{cmd}}\, \Delta t},\ -12,\ +12 \right) \quad (w = 0.5)$$

$\hat{e}_h(k)$：clip 第 k 帧的啮合调度；e_h：实际 latch。高水位项 settle 0.5 s 内只跟随不计费；两项仅在 $|v_z^{\text{cmd}}| > 0.01$ 时激活。

注：上限 12 倍参考速度且无逐 episode 预算帽——motionref1 “撕爪狂奔” （height: 模仿收入 15:1）的直接原因；motionref2 起加 1.25× 参考里程预算帽。

3　姿态 / 几何护栏

$$r_{\text{lat}} = \min\Big(\big(v_y/0.1\big)^2,\ 4\Big) \quad (w = -0.2)$$

$$r_{\text{wall}} = \min\left(\left(\frac{\big(|d - 0.196| - 0.08\big)_+}{0.1}\right)^2,\ 4\right), \qquad d = \text{基座—墙面距离} \quad (w = -0.2)$$

$$r_{\text{face}} = \frac{1}{|\mathcal{M}|}\sum_{h\in\mathcal{M}}\big(1 - s_h\big), \qquad s_h = \tfrac{1}{2}\big(1 + \cos\gamma_h\big) \quad (w = -0.5)$$

γ_h：EE face 轴 $(0, -1, 0)$ 与墙内法线夹角；掩码 $\mathcal{M} = \{h : e_h \vee (\text{接触球数}_h > 0)\}$，悬空摆动肢不计。

4 Microspine 物理契约

$$r_{\text{support}} = \max\bigl(0,\ 2 - \textstyle\sum_h e_h\bigr) \quad (w = -4.0)$$

$$r_{\text{release}} = \frac{1}{\Delta t}\left[\sum_h \text{rel}_h\, c(\text{reason}_h) \ +\ \mathbf{1}\Bigl[\exists\,\text{rel} \wedge \textstyle\sum_h e_h < 2\Bigr]\cdot 1.0\right] \quad (w = -1.0)$$

$$c = \{\text{reverse} : 0,\ \text{tension} : 0.25,\ \text{lateral} : 0.25,\ \text{ultimate} : 1.0\}$$

$$r_{\text{slide}} = \sum_h e_h \left\|\boldsymbol{v}_{h,\parallel}\right\|, \qquad \boldsymbol{v}_{h,\parallel} = \boldsymbol{v}_h - (\boldsymbol{v}_h \cdot \boldsymbol{n})\,\boldsymbol{n} \quad (w = -0.1)$$

释放罚 $\div\Delta t$ 抵消 manager 的 $\times\Delta t \Rightarrow$ 每事件一次性成本；reverse（反耙脱钩）免费 = 预期的主动释放路径；settle $0.5\,\text{s}$ 免罚。

5　正则化

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\begin{aligned}
r_{\text{angvel}} &= \omega_x^2 + \omega_y^2 \quad \text{(基座体系，yaw 不罚)} \ \ (w = -0.05) \\
r_{\tau} &= \textstyle\sum_j \tau_j^2 \ \ (w = -10^{-5}) \\
r_{\text{acc}} &= \min\!\big(\text{mean}_j(\ddot{q}_j/100)^2,\ 10\big) \ \ (w = -0.02) \\
r_{\text{rate}} &= \textstyle\sum_j (a_{j,t} - a_{j,t-1})^2 \ \ (w = -0.01) \\
r_{\text{limit}} &= \textstyle\sum_j \left[(q_j^{\min} - q_j)_+ + (q_j - q_j^{\max})_+\right] \ \ (w = -5.0) \\
r_{\text{term}} &= \mathbf{1}[\text{非超时终止}] \ \ (w = -500)
\end{aligned}
$$

6 Actor 观测（单帧，无噪声）

$$\boldsymbol{o}^{\text{actor}} = \Big[\underbrace{\boldsymbol{v}^b,\ \boldsymbol{\omega}^b,\ \boldsymbol{g}^b}_{\text{传感器}3\times3},\ \underbrace{\boldsymbol{q}-\boldsymbol{q}^{\text{def}},\ \dot{\boldsymbol{q}}}_{2n_j},\ \underbrace{\boldsymbol{a}_{t-1},\ \boldsymbol{a}_{t-2},\ \boldsymbol{a}_{t-3}}_{3n_j},\ z-2.0,\ \boldsymbol{v}^{\text{cmd}},\ \boldsymbol{o}^{\text{spine}},\ \boldsymbol{o}^{\text{motion}}\Big]$$

$$\boldsymbol{o}^{\text{spine}} = \Big[e_h,\ \ \text{clip}\big(\tfrac{\#\text{contact}_h}{4},0,1\big),\ \ \tfrac{\min(t_h^{\text{dis}},1.5)}{1.5},\ \ \tfrac{\text{cooldown}_h}{10}\Big]_{h=1..4} \in \mathbb{R}^{16}$$

$$\boldsymbol{o}^{\text{motion}} = \big[\cos\varphi,\ \sin\varphi,\ \hat{e}_1,\ldots,\hat{e}_4\big] \in \mathbb{R}^6, \qquad \varphi = 2\pi k/K \ \ \text{(循环相位钟)}$$

7 Critic 特权观测（非对称 AC：critic = actor ∪ 下列）

$$m_h = \min\left(\frac{f_t + f_{\text{rev}}}{f_{\text{rev}}},\ \frac{c_n - f_n}{c_n},\ \frac{c_l - |f_l|}{c_l},\ \frac{f_t^{\max} - f_t}{f_t^{\max}}\right) \quad (\text{wedge 四面裕度，未啮合} = 1)$$

$$c_n = \mu_{\text{adh}} f_t^{+} + f_{n0}, \qquad c_l = \mu_{\text{lat}} f_t^{+} + f_{l0}$$

$$\boldsymbol{o}^{\text{load}} = 0.01 \cdot \left[f_t, f_l, f_n\right]_{h=1..4} \in \mathbb{R}^{12} \quad (\text{锚点载荷，啮合系})$$

$$\boldsymbol{o}^{\text{rel}} = \left[\text{rel}_h\right]_{h=1..4} \in \{0,1\}^4 \quad (\text{本步 limit-surface 强制释放})$$

本版 limit surface 为 wedge 线性锥（wang2017 闭合曲面 07-23 起才替换，届时 m_h 语义改为径向裕度 $\frac{\text{cap}-\|F\|}{\text{cap}+\epsilon F_0}$）。